Lab Task 1

Task scenario

A small shop needs an system to manage its products. The shop must be able to add products, view all products, find one product by ID, update product details, and delete a product.

Product fields

Field	Type	Description
id	Generated ID	Primary key generated by the database
name	String	Product name
category	String	Product category
price	Number	Product price
quantity	Integer	Available stock quantity

Required endpoints

Method	Endpoint	Required behavior
POST	/products	Create and save a product
GET	/products	Return all saved products
GET	/products/:id	Return the product with the given ID
PATCH	/products/:id	Update the product with the given ID
DELETE	/products/:id	Delete the product with the given ID

Employee fields

Field	Type	Description
id	Generated ID	Primary key generated by the database
name	String	Employee full name
email	String	Employee email address
phone	String	Phone number
password	String	Dummy password
age	Integer	Employee age in years
yearsOfWorking	Integer	Number of completed years of work experience

Required endpoints

Method	Endpoint	Required behavior
POST	/employees	Create and save an employee
GET	/employees	Return all saved employees
GET	/employees/:id	Return the employee with the given ID
PATCH	/employees/:id	Update the employee with the given ID
DELETE	/employees/:id	Delete the employee with the given ID

Implementation instructions

- Use a product module, controller, service, and entity or model. Keep database operations in the service.
- Use the database and ORM or database library taught in class. Read and write actual database records; an in-memory array is not sufficient.

GitHub folder structure

Upload your work inside your main course repository using this exact order:

Main Repo → Mid → Lab task 1 → product-management

Main Repo means your existing course repository. Place the complete NestJS project inside product-management.

Location inside product-management-api	Required contents
src/	NestJS application and product module files
screenshots/	Postman screenshots for all required operations
README.md	**Student details [SL, Name and ID]**, setup instructions
package.json and lockfile	Dependencies and commands needed to run the app
.gitignore and .env.example	Ignored local files and sample database settings
Other generated project files	NestJS and TypeScript configuration files

Exclude node_modules, dist, and your real .env file. Provide sample configuration values in .env.example.

Postman screenshots

Create at least two products. Save clear screenshots of successful requests in the screenshots folder using these names:

Filename	Evidence to show
01-create.png	POST request body and the saved product response
02-get-all.png	GET response showing at least two products
03-get-by-id.png	GET response for one actual product ID
04-update.png	PATCH request body and successful response
05-confirm-update.png	GET by ID showing the changed values
06-delete.png	DELETE request and successful status or response
07-confirm-delete.png	GET all showing the deleted product is absent

Each screenshot must show the HTTP method, full request URL, response status, and response body when present. Show the JSON request body for POST and PATCH. Keep text readable and do not expose database passwords.

Submission instructions

1. Add your name, student ID, and section to README.md. Include database setup steps, installation and run commands, and the endpoint list.
2. Commit and push the code and screenshots to GitHub. Ensure the instructor can access the repository.
3. Submit the GitHub link pointing directly to Mid/Lab task 1/product-management-api through the course submission channel.

Before submitting: confirm that the database is connected, every CRUD operation works, all screenshots are uploaded, and the latest commit is visible on GitHub.

Submission deadline: 20 October 2026 at 11:59 AM.